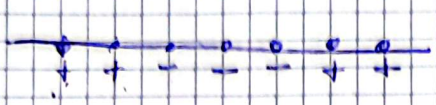
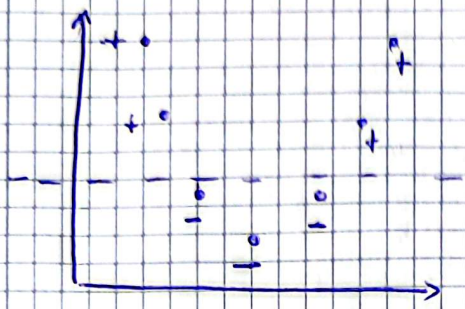
Exercise 1a

1- 

Yes it is possible

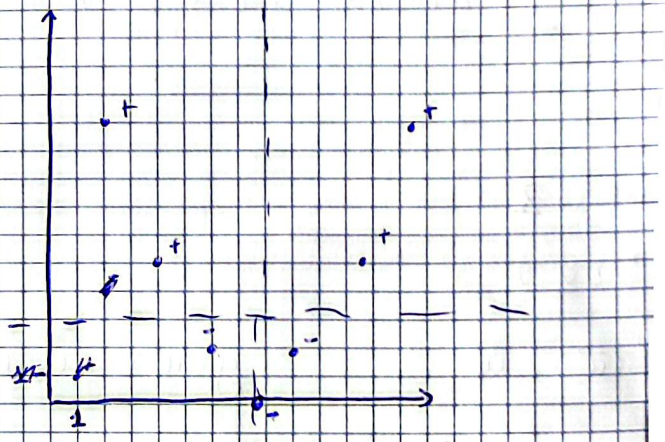


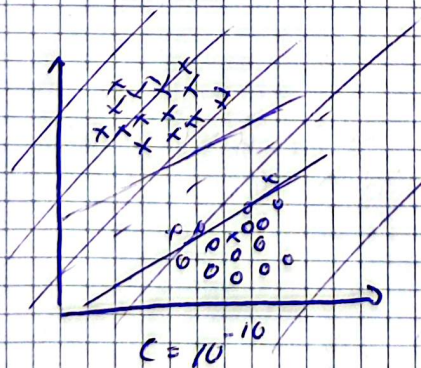
$\phi(x) = (x, x^2)$

with this

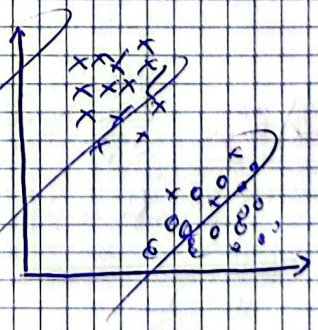
Let's give some values

x	x^2
+ -3	9
+ -2	4
- -1	1
- 0	0
- 1	1
+ 2	4
+ 3	9



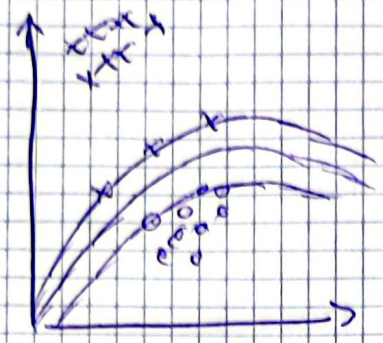
2- 

$C = 10^{-10}$

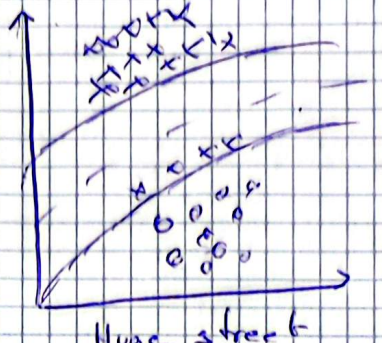


Escaneado con CamScanner

Part b



Narrow street
$C = 10^{10}$



Huge street
$C = 10^{-10}$

Since C defines how costly is to break the conditions (how accurate the prediction has to be) with a small $C = 10^{-10}$ this means that is more important to maximize the width of $\vec{u}$ will over accuracy while it's the opposite for $C = 10^{10}$

Exercise 3

$\vec{v} = [6 \ 9 \ 3]^T$

$f(\vec{u}) = \vec{u} \cdot \vec{v}$ $\quad g(\vec{u}) = \vec{u} \cdot \vec{u} - 1 = 0$

$\mathcal{L} = f(\vec{u}) - \lambda g(\vec{u})$

$\mathcal{L} = \vec{u} \cdot \vec{v} - \lambda(\vec{u} \cdot \vec{u} - 1)$

$\frac{\partial \mathcal{L}}{\partial \vec{u}} = \vec{v} - 2\lambda\vec{u} = 0$ $\quad \frac{\partial \mathcal{L}}{\partial \vec{v}} \vec{u} = 0$ No variable $\quad \frac{\partial \mathcal{L}}{\partial \lambda} = \vec{u} \cdot \vec{u} - 1 = 0$

$\vec{v} - 2\lambda\vec{u} = 0 \ ; \ \vec{u} = \frac{\vec{v}}{2\lambda}$

$\vec{u} \cdot \vec{u} = 1 \rightarrow \frac{\vec{v}}{2\lambda} \cdot \frac{\vec{v}}{2\lambda} = 1$

$\frac{1}{2\lambda^2}(\vec{v} \cdot \vec{v}) = 1 \rightarrow \frac{1}{4\lambda^2}(36 + 81 + 9) = 1$

$\frac{1}{4\lambda^2} 126 = 1 \ ; \ 4\lambda^2 = 126 \ \lambda = \sqrt{126/4}$

$\lambda^2 = 126/4 \ \lambda = \pm 5,61$

Escaneado con CamScanner

$\vec{u} = \frac{\vec{v}}{2\Delta}$

$\rightarrow \frac{[6\ 2\ 3]}{2 \cdot \sqrt{5,61}} = [16,83 ; 25,245 ; 8,415]$

$\rightarrow \frac{-[6\ 2\ 3]}{2\ 5,61} = [-16,83 ; -25,245 ; -8,415]$

$\vec{u} = \frac{\vec{v}}{2\Delta}$

$\rightarrow \frac{[6\ 2\ 3]}{2 \cdot 5,61} = [0,5342 \ \ 0,8021 \ \ 0,2674] \rightarrow 11,22$

$\rightarrow \frac{-[6\ 2\ 3]}{2 \cdot 5,61} = [-0,5342 \ \ -0,8021 \ \ -0,2674] \rightarrow -11$

$$\boxed{\vec{u} = \frac{[6\ 2\ 3]}{2\sqrt{\frac{126}{4}}}}$$

Exercise 3

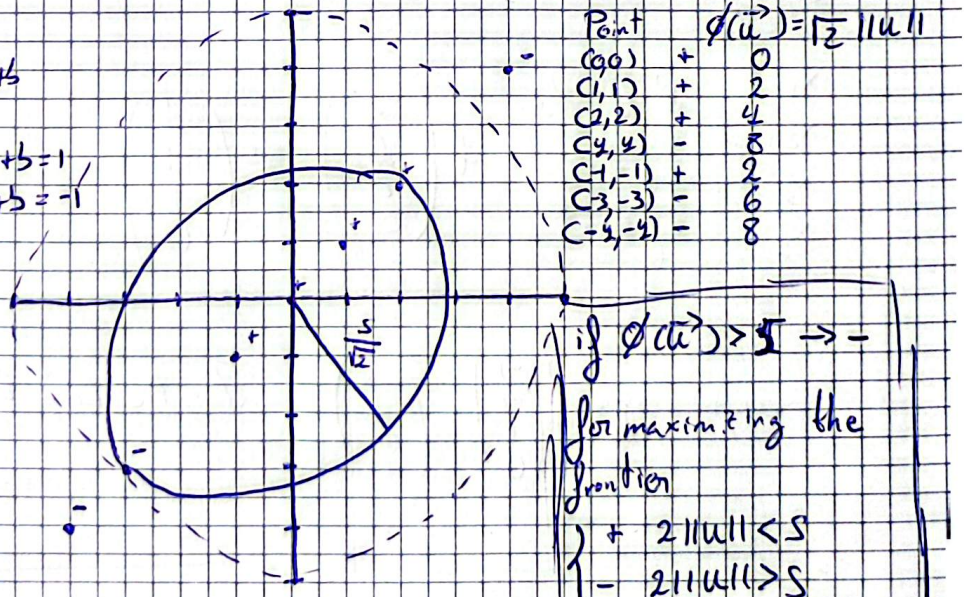
1-

$f(\phi) = w\phi + b$

positive $\begin{cases} 4w + b = 1 \\ 6w + b = -1 \end{cases}$

$w = -2$

$\boxed{b = 9}$



Point		$\phi(\vec{u}) = \sqrt{2}\|u\|$
(0,0)	+	0
(1,1)	+	2
(2,2)	+	4
(4,4)	-	8
(-1,-1)	+	2
(-3,-3)	-	6
(-4,-4)	-	8

if $\phi(\vec{u}) > 5 \rightarrow -$

for maximizing the frontier

$\begin{cases} + & 2\|u\| < 5 \\ - & 2\|u\| > 5 \end{cases}$

$K(\vec{u}, \vec{v}) = \phi(\vec{u}) \cdot \phi(\vec{v})$

$\phi(\vec{v})$ has to only depend on $\vec{v}$ but we can multiply it. I propose $\phi(\vec{v}) = c\|\vec{v}\|$

$K(\vec{u}, \vec{v}) = 2|\vec{u}||\vec{v}|$

$\phi(\vec{u}) \cdot \phi(\vec{v}) = 2|\vec{u}||\vec{v}|$

$c|\vec{u}| \cdot c|\vec{v}| = 2|\vec{u}||\vec{v}|$

$c^2 = 2 \ ; \ c = \sqrt{2}$

Escaneado con CamScanner

2- $A = \begin{bmatrix} 1 & 1 & 2 & 1 \\ 3 & 1 & 4 & 4 \\ 4 & -4 & 0 & 8 \end{bmatrix}$

$\ker(A) := \{x \in \mathbb{R}^4 \mid Ax = 0\}$

$$\begin{pmatrix} 1 & 1 & 2 & 1 \\ 3 & 1 & 4 & 4 \\ 4 & -4 & 0 & 8 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 1 & 2 & 1 \\ 0 & -2 & -2 & 1 \\ 0 & -8 & -8 & -4 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 1 & 2 & 1 \\ 0 & -2 & -2 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$\begin{cases} x_1 + x_2 + 2x_3 + x_4 = 0 \\ -2x_2 - 2x_3 + x_4 = 0 \end{cases}$ $\quad x_3 = s \quad x_4 = t$

$2x_4 + x_1 - x_2 = 0 \quad \Rightarrow x_1 = x_2 - 2x_4$

~~$x_2 = x_1 = \dfrac{t - 2s}{2}$~~

$x_2 = \dfrac{t-2s}{2} \qquad x_1 = \dfrac{t-2s}{2} + 2t \qquad x_1 = \dfrac{3t+2s}{2} = -\left(\dfrac{3t+2s}{2}\right)$

$$\begin{matrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{matrix} \quad \begin{matrix} -\left(\frac{3t+2s}{2}\right) \\ \frac{t-2s}{2} \\ s \\ t \end{matrix} = t\begin{pmatrix} -\frac{3}{2} \\ \frac{1}{2} \\ 0 \\ 1 \end{pmatrix} + s\begin{pmatrix} -1 \\ -1 \\ 1 \\ 0 \end{pmatrix}$$

$\ker(A) = \operatorname{span}\left\{\begin{pmatrix} -1 \\ -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -3 \\ 1 \\ 0 \\ 2 \end{pmatrix}\right\} \qquad \dim \ker(A) = 2$

Escaneado con CamScanner

3- $K(x,z) = (xz+2)^2$ Has to be symetrice and positive semi-definied

$K(x,z) = (xz)^2 + 4xz + 4$

$K(x,z) = \phi(x)\phi(z)$

$\phi(a) = (a^2, ca, c)$

$\phi(x)\phi(z) = x^2z^2 + c^2xz + c^2 \Big|_{c=2} = K(x,z)$

It is a valid Kernel